GG-B-95E
30 May 1986
SUPERSEDING
GG-B-0095D (DSA-DM)
7 June 1972
Interim Revision of
GG-B-95C
10 October 1962

FEDERAL SPECIFICATION

BAND AND BAND SET, MATRIX, DENTAL, Tofflemire

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers corrosion-resisting steel matrix bands suitable for use with matrix retainers for restorations in individual teeth.

1.2 Classification.

1.2.1 Class, size and thickness. Bands shall be of the following class, size and thickness, as specified (see 3.5 and 6.2).

Class 1. Bands, Junior size, Tofflemire.

Size (number)
Universal, No. 13
Narrow M.O.D., No. 14
Wide M.O.D., No. 15

Class 2. Bands, Adult size, Tofflemire.

Size (number)
Universal, No. 1
Wide M.O.D., No. 2
Narrow M.O.D., No. 3
Window-Opening Contra Angle, No. 10
Universal Contra Angle, No. 11

AMSC N/A FSC 6520

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Thickness (inches)
0.0015
0.0020

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications

GG-I-526	Instruments, Dental and Surgical, General Specifications.
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-601	Boxes, Wood, Cleated-plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	Boxes, Shipping, Fiberboard.
PPP-B-676	Boxes, Set-Up.

Federal Standards

FED-STD-123	Marking for Domestic Shipment (Civilian Agencies).
FED-STD-151	Metal, Test Methods.

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

(Copies of listed federal and military standards, specifications, Commercial Item Descriptions (CIDS), handbooks and associated documents listed in the Department of Defense Index of Specifications and Standards (DoDISS), should be obtained from the DoD Single Stock Point, Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120. Copies of industry association documents should be obtained from the sponsor. Copies of all other listed documents should be obtained from the contracting activity or as directed by the contracting activity.

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

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Military Specifications

MIL-P-116	Preservation, Methods Of.

Military Standards

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-147	Palletized Unit Loads.
MIL-STD-794	Parts and Equipment, Procedures for Packaging Of.

DRAWING

DEFENSE PERSONNEL SUPPORT CENTER (DPSC)

23355	Band, Matrix, Dental, Tofflemire, Adult and Junior Sizes.

(Copies of Military specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents forms a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 18	Tests for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Material.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Drawing. The drawing forming a part of this specification is a specification control drawing. The supplier is responsible for preparing his own shop drawing. When tolerances prescribed could cumulatively result in improper design, improper function or result in the band being unsuitable for use with the applicable retainer (as specified in 1.2), the supplier shall utilize tolerances within those prescribed which will result in proper design, function, and use. No deviation from the prescribed dimensions or tolerances is permissible without prior approval from the contracting officer.

3.2 Materials. Bands shall be fabricated of corrosion resisting steel conforming to the requirements for AISI 301, 302, or 304 of GG-I-526.

3.3 Hardness. Bands shall have a Knoop hardness number of 351 to 402 (Rockwell C35 to C40) when tested as specified in 4.4.2.

3.4 Style, design and dimensions. Bands shall be suitable for use with the applicable retainers specified in paragraph 3.4.1.1 and 3.4.1.2. Matrix bands shall be capable of being fitted around a tooth to supply a missing wall or contour, against which the filling material may be placed and condensed.

3.4.1 Bands. Bands shall be furnished in pre-stamped designs conforming to the applicable class and size (number) specified.

3.4.1.1 Class 1. Class 1 bands shall be suitable for use with junior size Tofflemire retainers and shall conform to the design, dimensions, and size (number) shown in DPSC Drawing No. 23355.

3.4.1.2 Class 2. Class 2 bands shall be suitable for use with adult size Tofflemire matrix retainer and shall conform to the design, dimensions, and size (number) shown in DPSC Drawing No. 23355.

3.4.2 Thickness. Bands shall conform to the thickness specified for the type, class, and size number specified in Table 1.

3.5 Class, size and thickness. Bands shall be furnished in the applicable class, size number, thickness and quantity shown in Table I.

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Table I. Class, Size (number), Thickness and Quantity.

Class	Size (number)	Thickness, inches 1/	Quantity	Total Quantity	Unit of Issue
Bands					
1. Junior Size Tofflemire	Universal #13	0.0020	6		Set
	Narrow MOD #14		2	12	
	Wide MOD #15		4		
2. Adult Size Tofflemire	Universal #1	0.0015	12	12	Package
	Universal #1	0.0020	12	12	Package
	Wide MOD #2	0.0015	12	12	Package
	Narrow MOD #3	0.0015	12	12	Package
	Window-Opening Contra-Angle, #10	0.0015	4		
	Universal Contra-Angle #11	0.0015	8	12	Set

1/ Thickness tolerance for 0.0020 ± 0.0002 inch
0.0015 ± 0.0002 inch

3.6 Corrosion resistance. Bands shall show no signs of corrosion when subjected to the boil test specified in 4.4.3.

3.7 Flexibility. Bands shall withstand a minimum of two bends before breaking, when tested as specified in 4.4.4. A bend shall consist of one flattening followed by straightening.

3.8 Finish. The bands shall have a commercial number 2B finish or better. The bands shall be cleanly trimmed and smoothly finished. All bands shall be put through a tumbling procedure to provide smooth peripheral edges free from burrs or other imperfections. The surface roughness of bands shall be no more than the root-mean-square average deviation of 15 microinches when tested as specified in 4.4.5.

3.9 Workmanship. The finish of the bands shall be uniform and free of burrs, sharp edges, pit marks, cracks, wrinkles, folds, tears or other imperfections. The matrix bands shall be free from any imperfections which may detract from their appearance or impair their serviceability, proper functioning, and intended use.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Quality conformance inspection. The examination and tests required to assure conformance of those items or lots of items to be offered are classified as specified in Tables II and IV (see 4.2.1 and 4.3) and Table III (see 4.2.2).

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) shall be as indicated in Table II (see 4.2.3).

Table II. Examination Critieria

Type defect	Inspection level	AQL (Percent defective
Major A	II	1.0
Major B	S-1	2.5
Minor	II	4.0

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4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table III except that the acceptance number shall be zero (0) for all sample sizes. Where feasible, the same unit shall be used for the determination of two or more test characteristics (see 4.2.3).

Table III. Tests

Characteristic	Requirement	Test procedure	Inspection level
Chemical composition	3.2	4.4.1	See 4.2.2.1
Hardness (Finished item)	3.3	4.4.2	S-2*
Corrosion resistance (Finished item)	3.6	4.4.3	S-2*
Flexibility (Finished item)	3.7	4.4.4	S-2*
Surface roughness (Finished item)	3.8	4.4.5	S-2*

*Three 3-inch long specimens cut from class 1 and class 2 bands shall be tested from each unit of product.

4.2.2.1 Sampling for raw materials. A sufficient amount from each lot of metal used in the fabrication of the bands shall be selected for chemical analysis.

4.2.2.2 Lots for testing critieria. For finished item testing, a lot or batch shall consist of finished items, produced by one manufacturer, at one plant, and from the same material (same raw material heat).

4.2.3 Unit of product. The unit of product for examination and test purposes shall be one unit as specified in Table I for the applicable class and size.

4.3 Examination. The bands shall be examined to determine compliance with specification requirements. Examination shall be conducted in accordance with 4.2.1. The bands shall be examined for defects, including but not limited to, those listed in Table IV.

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Table IV. Classification of Defects.

Categories	Defects
Major A:	
101	Class, size (number) and thickness not as specified.
102	Edges not free of burrs or other imperfections.
103	Finish not uniform.
104	Finish not free of scratches, pit marks or cracks
105	Bands not free of wrinkles, folds, breaks or tears.
106	Bands not free from dirt, grease or other foreign matter.
107	Design not as specified.
Major B:	Dimensions not as specified
Minor:	
201	Bands cutout not cleanly stamped, or distorted in design.

4.4 Tests. Except for chemical composition, tests shall be conducted on finished bands to determine compliance with the requirements of this specification (see 4.2.2).

4.4.1 Chemical composition. In the absence of certificates of quality for the metallic raw material, chemical analysis shall be conducted in accordance with method 111 or 112 of FED-STD-151. In case of dispute, method 111 shall govern.

4.4.2 Hardness. Hardness shall be determined in accordance with ASTM Designation E18 (Rockwell). Knoop hardness shall be determined on a Tukon tester under a 500-gram load, using a rhombus-shaped diamond indenting tool with a longitudinal angle of 172.5 degrees and a traverse angle of 130 degrees.

4.4.3 Corrosion resistance. Bands shall be tested for corrosion resistance in accordance with the requirements of GG-I-526.

4.4.4 Flexibility test. The matrix shall be bent back on itself by creasing it in the middle (so that one half is superimposed on the other) and returning it to its original flat position. The flattening and straightening of the 180 degree bend shall be done using a rubber-covered metal roller weighing 4-1/2 plus or minus 1/2 pounds. Subsequent bends shall be performed in the same way with the inside surface of the crease the same for all bends.

4.4.5 Surface roughness test. A suitable surface roughness analyzer shall be used to determine the surface roughness of the matrix bands.

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4.5 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Bands shall be cleaned and dried in accordance with MIL-P-116.

5.1.1.2 Preservative applications. Preservatives shall not be used.

5.1.1.3 Unit package. One package or one set containing twelve bands of the class and size as specified in table I shall be packaged in accordance with MIL-P-116, method III in a sealed envelope of an appropriate size and design.

5.1.1.4 Intermediate package. One hundred unit packages containing bands of one class and size only shall be packaged in a box conforming to PPP-B-566, PPP-B-636, class domestic or PPP-B-676. Closure shall be as specified in the applicable box specification.

5.1.2 Level C. The level C preservation for bands shall conform to the MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. Thirty six intermediate packages containing bands of one class and size only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, overseas type, PPP-B-621, class 2 or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall be as specified in the liner specification. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in PPP-B-636.

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5.2.2 Level B. Thirty-six intermediate packages containing bands of one class and size only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, domestic type, PPP-B-621, class 1 or PPP-B-636, class domestic. Closure shall be as specified in the applicable box specification.

5.2.3 Level C. The level C packing for bands shall conform to the MIL-STD-794 requirements for this level.

5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Loads shall be unitized on pallets conforming to MIL-STD-147. Quantities less than 40 cubic feet not be unitized. Pallet loads including the pallet shall not exceed 43 inches in height, 48 inches in width, and 40 inches in length, and shall conform to Type IV or Type V.

5.2.5.1 Levels A, B and C. Bands packed as specified in 5.2.1, 5.2.2, and 5.2.3 shall be unitized on pallets as specified in 5.2.5.

5.3 Marking.

5.3.1 Levels A, B, and C. Each unit pack, intermediate package, exterior container and unitized load shall be marked as specified in MIL-STD-129.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

6. NOTES

6.1 Intended use. Bands covered by this specification are intended for use with applicable retainers to supply a missing wall or contour during dental procedures.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Class, size (number) and thickness required (see 1.2.1 and 3.5).
(c) Selection of applicable levels of packaging and packing.

6.3 National Stock Number coverage. The following National Stock Numbers are covered by this document:

Stock Number	Item Identification
6520-00-682-6530	BAND SET, MATRIX, DENTAL, Assorted, Tofflemire, Junior Sizes, 0.0020 in. thick.
6520-00-721-9351	BAND, MATRIX, DENTAL, Tofflemire, Adult Size, No. 3, 0.0015 in. thick
6520-00-721-9355	BAND, MATRIX, DENTAL, Tofflemire, Adult Size, No. 1, 0.0015 in. thick.
6520-00-721-9356	BAND, MATRIX, DENTAL, Tofflemire, Adult Size, No. 2, 0.0015 in. thick.
6520-00-817-2297	BAND SET, MATRIX, DENTAL, Assorted, Tofflemire, Adult Size, No. 10 and 11, 0.0015 in. thick.
6520-01-079-6363	BAND, MATRIX, DENTAL, Tofflemire, Adult Size, No. 1, 0.0020 in. thick.

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air-Force - 03

Review Activities:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD-MB

AGENT:

DLA-DM

Civil Agency Coordinating Agency:
VA-OSS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
GG-B-95E	BAND AND BAND SET, MATRIX, DENTAL, TOFFLEMIRE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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